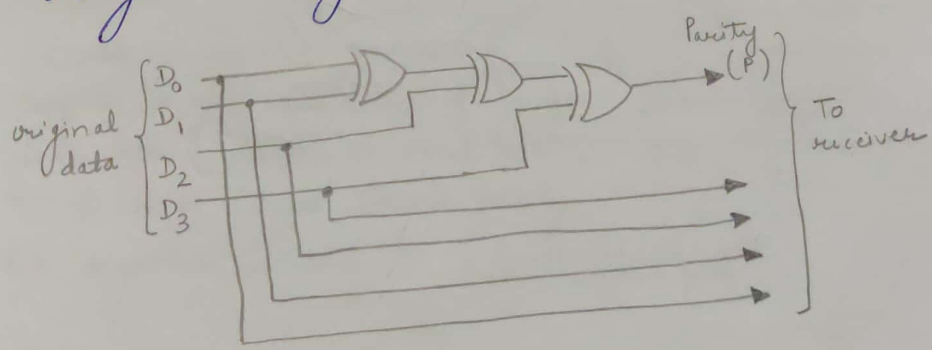
Parity Bit Generators / Checkers

1. X-OR functions are very useful in systems requiring error detection & correction codes.
2. We have already discussed parity check method - Binary data is susceptible to errors. To detect such errors, parity bit is added to the data bits & word containing both data & parity bit is transmitted. A parity bit, 0 or 1 is attached such that total no. of 1s is even for even parity & odd for odd parity.

   At the receiving end, if word has received an even no. of 1s in odd parity system & or an odd no. of 1s in even parity system, an error has occurred.
3. The circuit that generates the parity bit in the transmitter is called a parity generator.
4. The circuit that checks the parity in the receiver is called a parity checker.
5. The basic principle used in order to generate/check parity is " the modulo sum (XOR) of an even no. of 1s is always a 0 & modulo sum of an odd no. of 1s is always a 1."

Even parity generator -

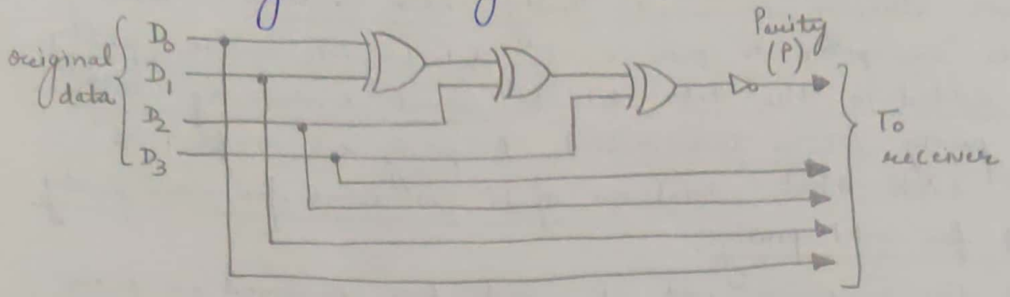
To generate an even parity bit, four data bits are added using 3 X-OR gates. The sum bit will be parity bit.



Even parity generator

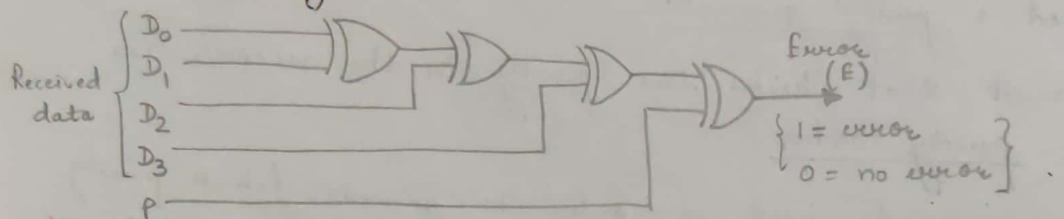
Odd parity generator -

To generate an odd parity bit, the four data bits are added using 3 X-OR gates & the sum bit is inverted.



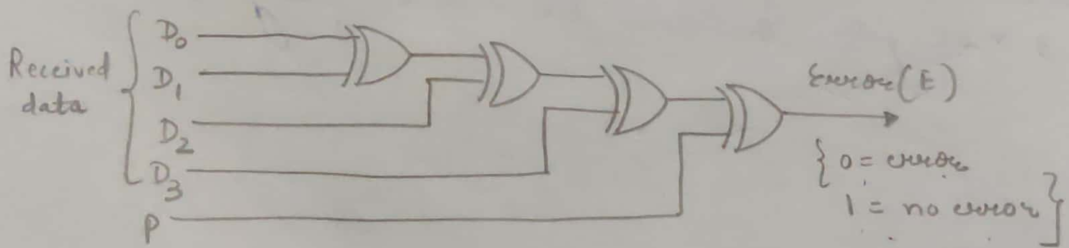
Odd parity generator

Even bit parity checker -



Even parity checker

Odd parity checker -

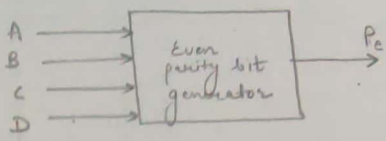


Odd parity checker

Scanned by CamScanner

Design of an even parity bit generator for a 4-bit i/p -

Let the 4-bit i/p be ABCD.
For even parity, parity bit 1 is added such that total no. of 1's in 4-bit i/p & parity bit together is even.



4-bit data i/p A	B	C	D	Output parity bit, P_e
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

AB \ CD	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

K-map for f

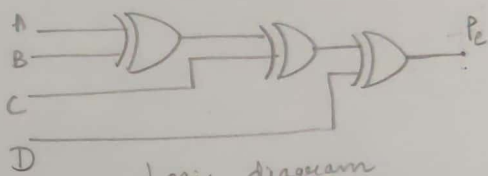
$$P_e = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + AB\bar{C}D + ABC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}CD$$

$$= \bar{A}\bar{B}(C\oplus D) + \bar{A}B(\overline{C\oplus D}) + AB(C\oplus D) + A\bar{B}(\overline{C\oplus D})$$

$$= (C\oplus D)(\overline{A\oplus B}) + (\overline{C\oplus D})(A\oplus B)$$

$$\boxed{P_e = A\oplus B\oplus C\oplus D}$$

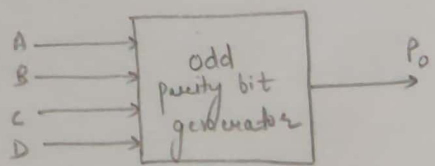
$\Downarrow$



Logic diagram

Design of an odd parity bit generator for a 4-bit i/p -

An odd parity bit generator outputs a 1, when no. of 1s in data bits is even, so that total no. of 1s in data bits & parity bit together is odd.



4-bit data i/p				Output parity bit, P_0
A	B	C	D	
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

AB \ CD	00	01	11	10
00	1		1	
01		1		1
11	1		1	
10		1		1

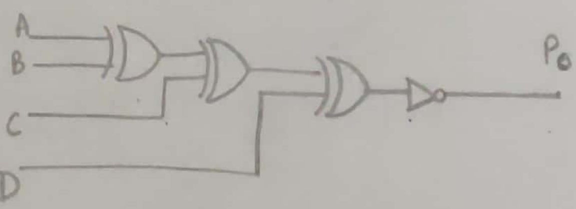
$$P_0 = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + AB\bar{C}\bar{D} + ABCD + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D}$$

$$= \bar{A}\bar{B}(\overline{C\oplus D}) + AB(\overline{C\oplus D}) + \bar{A}B(C\oplus D) + A\bar{B}(C\oplus D)$$

$$= (\overline{C\oplus D})(\overline{A\oplus B}) + (C\oplus D)(A\oplus B)$$

$$= \overline{A\oplus B\oplus C\oplus D}$$

$$\Downarrow$$



Logic diagram